

USAJMO

§1 USAJMO 2010

1. (*USAJMO 2010*) A permutation of the set of positive integers $[n] = \{1, 2, \dots, n\}$ is a sequence $(a_1, a_2, \dots, a_n)$ such that each element of $[n]$ appears precisely one time as a term of the sequence. For example, $(3, 5, 1, 2, 4)$ is a permutation of $[5]$. Let $P(n)$ be the number of permutations of $[n]$ for which ka_k is a perfect square for all $1 \leq k \leq n$. Find with proof the smallest n such that $P(n)$ is a multiple of 2010.
2. (*USAJMO 2010*) Let $n > 1$ be an integer. Find, with proof, all sequences $x_1, x_2, \dots, x_{n-1}$ of positive integers with the following three properties:
 - a) $x_1 < x_2 < \dots < x_{n-1}$;
 - b) $x_i + x_{n-i} = 2n$ for all $i = 1, 2, \dots, n-1$;
 - c) given any two indices i and j (not necessarily distinct) for which $x_i + x_j < 2n$, there is an index k such that $x_i + x_j = x_k$.
3. (*USAJMO 2010*) Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB .
4. (*USAJMO 2010*) A triangle is called a *parabolic triangle* if its vertices lie on a parabola $y = x^2$. Prove that for every nonnegative integer n , there is an odd number m and a parabolic triangle with vertices at three distinct points with integer coordinates with area $(2^n m)^2$.
5. (*USAJMO 2010*) Two permutations $a_1, a_2, \dots, a_{2010}$ and $b_1, b_2, \dots, b_{2010}$ of the numbers $1, 2, \dots, 2010$ are said to intersect if $a_k = b_k$ for some value of k in the range $1 \leq k \leq 2010$. Show that there exist 1006 permutations of the numbers $1, 2, \dots, 2010$ such that any other such permutation is guaranteed to intersect at least one of these 1006 permutations.
6. (*USAJMO 2010*) Let ABC be a triangle with $\angle A = 90^\circ$. Points D and E lie on sides AC and AB , respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I . Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer lengths.

§2 USAJMO 2011

1. (*USAJMO 2011*) Find, with proof, all positive integers n for which $2^n + 12^n + 2011^n$ is a perfect square.

2. (USAJMO 2011) Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$. Prove that

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ca+1}{(c+a)^2} \geq 3.$$

3. (USAJMO 2011) For a point $P = (a, a^2)$ in the coordinate plane, let $\ell(P)$ denote the line passing through P with slope $2a$. Consider the set of triangles with vertices of the form $P_1 = (a_1, a_1^2)$, $P_2 = (a_2, a_2^2)$, $P_3 = (a_3, a_3^2)$, such that the intersections of the lines $\ell(P_1)$, $\ell(P_2)$, $\ell(P_3)$ form an equilateral triangle Δ . Find the locus of the center of Δ as $P_1P_2P_3$ ranges over all such triangles.
4. (USAJMO 2011) A *word* is defined as any finite string of letters. A word is a *palindrome* if it reads the same backwards as forwards. Let a sequence of words $W_0, W_1, W_2, \dots$ be defined as follows: $W_0 = a$, $W_1 = b$, and for $n \geq 2$, W_n is the word formed by writing W_{n-2} followed by W_{n-1} . Prove that for any $n \geq 1$, the word formed by writing $W_1, W_2, \dots, W_n$ in succession is a palindrome.
5. (USAJMO 2011) Points A, B, C, D, E lie on a circle ω and point P lies outside the circle. The given points are such that (i) lines PB and PD are tangent to ω , (ii) P, A, C are collinear, and (iii) $\overline{DE} \parallel \overline{AC}$. Prove that $\overline{BE}$ bisects $\overline{AC}$.
6. (USAJMO 2011) Consider the assertion that for each positive integer $n \geq 2$, the remainder upon dividing 2^{2^n} by $2^n - 1$ is a power of 4. Either prove the assertion or find (with proof) a counterexample.

§3 USAJMO 2012

1. (USAJMO 2012) Given a triangle ABC , let P and Q be points on segments $\overline{AB}$ and $\overline{AC}$, respectively, such that $AP = AQ$. Let S and R be distinct points on segment $\overline{BC}$ such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic (in other words, these four points lie on a circle).
2. (USAJMO 2012) Find all integers $n \geq 3$ such that among any n positive real numbers $a_1, a_2, \dots, a_n$ with

$$\max(a_1, a_2, \dots, a_n) \leq n \cdot \min(a_1, a_2, \dots, a_n),$$

there exist three that are the side lengths of an acute triangle.

3. (USAJMO 2012) Let a, b, c be positive real numbers. Prove that

$$\frac{a^3 + 3b^3}{5a + b} + \frac{b^3 + 3c^3}{5b + c} + \frac{c^3 + 3a^3}{5c + a} \geq \frac{2}{3}(a^2 + b^2 + c^2).$$

4. (USAJMO 2012) Let α be an irrational number with $0 < \alpha < 1$, and draw a circle in the plane whose circumference has length 1. Given any integer $n \geq 3$, define a sequence of points $P_1, P_2, \dots, P_n$ as follows. First select any point P_1 on the circle, and for $2 \leq k \leq n$ define P_k as the point on the circle for which the length of arc $P_{k-1}P_k$ is α , when travelling counterclockwise around the circle from P_{k-1} to P_k . Suppose that P_a and P_b are the nearest adjacent points on either side of P_n . Prove that $a + b \leq n$.

5. (*USAJMO 2012*) For distinct positive integers $a, b < 2012$, define $f(a, b)$ to be the number of integers k with $1 \leq k < 2012$ such that the remainder when ak divided by 2012 is greater than that of bk divided by 2012. Let S be the minimum value of $f(a, b)$, where a and b range over all pairs of distinct positive integers less than 2012. Determine S .
6. (*USAJMO 2012*) Let P be a point in the plane of triangle ABC , and γ a line passing through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to γ intersect lines BC, AC, AB , respectively. Prove that A', B', C' are collinear.

§4 USAJMO 2013

1. (*USAJMO 2013*) Are there integers a and b such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes of integers?
2. (*USAJMO 2013*) Each cell of an $m \times n$ board is filled with some nonnegative integer. Two numbers in the filling are said to be *adjacent* if their cells share a common side. (Note that two numbers in cells that share only a corner are not adjacent). The filling is called a *garden* if it satisfies the following two conditions:
 - (i) The difference between any two adjacent numbers is either 0 or 1.
 - (ii) If a number is less than or equal to all of its adjacent numbers, then it is equal to 0.

Determine the number of distinct gardens in terms of m and n .

3. (*USAJMO 2013*) In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z respectively, prove that $YX/XZ = BP/PC$.
4. (*USAJMO 2013*) Let $f(n)$ be the number of ways to write n as a sum of powers of 2, where we keep track of the order of the summation. For example, $f(4) = 6$ because 4 can be written as 4, $2 + 2$, $2 + 1 + 1$, $1 + 2 + 1$, $1 + 1 + 2$, and $1 + 1 + 1 + 1$. Find the smallest n greater than 2013 for which $f(n)$ is odd.
5. (*USAJMO 2013*) Quadrilateral $XABY$ is inscribed in the semicircle ω with diameter XY . Segments AY and BX meet at P . Point Z is the foot of the perpendicular from P to line XY . Point C lies on ω such that line XC is perpendicular to line AZ . Let Q be the intersection of segments AY and XC . Prove that

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{AY}{AX}.$$

6. (*USAJMO 2013*) Find all real numbers $x, y, z \geq 1$ satisfying

$$\min(\sqrt{x + xyz}, \sqrt{y + xyz}, \sqrt{z + xyz}) = \sqrt{x - 1} + \sqrt{y - 1} + \sqrt{z - 1}.$$

§5 USAJMO 2014

1. (*USAJMO 2014*) Let a, b, c be real numbers greater than or equal to 1. Prove that

$$\min\left(\frac{10a^2 - 5a + 1}{b^2 - 5b + 10}, \frac{10b^2 - 5b + 1}{c^2 - 5c + 10}, \frac{10c^2 - 5c + 1}{a^2 - 5a + 10}\right) \leq abc.$$

2. (*USAJMO 2014*) Let $\triangle ABC$ be a non-equilateral, acute triangle with $\angle A = 60^\circ$, and let O and H denote the circumcenter and orthocenter of $\triangle ABC$, respectively.
 - (a) Prove that line OH intersects both segments AB and AC .
 - (b) Line OH intersects segments AB and AC at P and Q , respectively. Denote by s and t the respective areas of triangle APQ and quadrilateral $BPQC$. Determine the range of possible values for s/t .
3. (*USAJMO 2014*) Let $\mathbb{Z}$ be the set of integers. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2f(2x - f(y)) = \frac{f(x)^2}{x} + f(yf(y))$$
 for all $x, y \in \mathbb{Z}$ with $x \neq 0$.
4. (*USAJMO 2014*) Let $b \geq 2$ be an integer, and let $s_b(n)$ denote the sum of the digits of n when it is written in base b . Show that there are infinitely many positive integers that cannot be represented in the form $n + s_b(n)$, where n is a positive integer.
5. (*USAJMO 2014*) Let k be a positive integer. Two players A and B play a game on an infinite grid of regular hexagons. Initially all the grid cells are empty. Then the players alternately take turns with A moving first. In his move, A may choose two adjacent hexagons in the grid which are empty and place a counter in both of them. In his move, B may choose any counter on the board and remove it. If at any time there are k consecutive grid cells in a line all of which contain a counter, A wins. Find the minimum value of k for which A cannot win in a finite number of moves, or prove that no such minimum value exists.
6. (*USAJMO 2014*) Let ABC be a triangle with incenter I , incircle γ and circumcircle Γ . Let M, N, P be the midpoints of sides $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ and let E, F be the tangency points of γ with $\overline{CA}$ and $\overline{AB}$, respectively. Let U, V be the intersections of line EF with line MN and line MP , respectively, and let X be the midpoint of arc BAC of Γ .
 - (a) Prove that I lies on ray CV .
 - (b) Prove that line XI bisects $\overline{UV}$.

§6 USAJMO 2015

1. (*USAJMO 2015*) Given a sequence of real numbers, a move consists of choosing two terms and replacing each with their arithmetic mean. Show that there exists a sequence of 2015 distinct real numbers such that after one initial move is applied to the sequence – no matter what move – there is always a way to continue with a finite sequence of moves so as to obtain in the end a constant sequence.
2. (*USAJMO 2015*) Solve in integers the equation

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1 \right)^3.$$

3. (*USAJMO 2015*) Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is

perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

4. (USAJMO 2015) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x) + f(t) = f(y) + f(z)$$

for all rational numbers $x < y < z < t$ that form an arithmetic progression. ($\mathbb{Q}$ is the set of all rational numbers.)

5. (USAJMO 2015) Let $ABCD$ be a cyclic quadrilateral. Prove that there exists a point X on segment $\overline{BD}$ such that $\angle BAC = \angle XAD$ and $\angle BCA = \angle XCD$ if and only if there exists a point Y on segment $\overline{AC}$ such that $\angle CBD = \angle YBA$ and $\angle CDB = \angle YDA$.
6. (USAJMO 2015) Steve is piling $m \geq 1$ indistinguishable stones on the squares of an $n \times n$ grid. Each square can have an arbitrarily high pile of stones. After he finished piling his stones in some manner, he can then perform stone moves, defined as follows. Consider any four grid squares, which are corners of a rectangle, i.e. in positions $(i, k), (i, l), (j, k), (j, l)$ for some $1 \leq i, j, k, l \leq n$, such that $i < j$ and $k < l$. A stone move consists of either removing one stone from each of (i, k) and (j, l) and moving them to (i, l) and (j, k) respectively, or removing one stone from each of (i, l) and (j, k) and moving them to (i, k) and (j, l) respectively.

Two ways of piling the stones are equivalent if they can be obtained from one another by a sequence of stone moves.

How many different non-equivalent ways can Steve pile the stones on the grid?

§7 USAJMO 2016

1. (USAJMO 2016) The isosceles triangle $\triangle ABC$, with $AB = AC$, is inscribed in the circle ω . Let P be a variable point on the arc $\widehat{BC}$ that does not contain A , and let I_B and I_C denote the incenters of triangles $\triangle ABP$ and $\triangle ACP$, respectively. Prove that as P varies, the circumcircle of triangle $\triangle PI_B I_C$ passes through a fixed point.
2. (USAJMO 2016) Prove that there exists a positive integer $n < 10^6$ such that 5^n has six consecutive zeros in its decimal representation.
3. (USAJMO 2016) Let $X_1, X_2, \dots, X_{100}$ be a sequence of mutually distinct nonempty subsets of a set S . Any two sets X_i and X_{i+1} are disjoint and their union is not the whole set S , that is, $X_i \cap X_{i+1} = \emptyset$ and $X_i \cup X_{i+1} \neq S$, for all $i \in \{1, \dots, 99\}$. Find the smallest possible number of elements in S .
4. (USAJMO 2016) Find, with proof, the least integer N such that if any 2016 elements are removed from the set $\{1, 2, \dots, N\}$, one can still find 2016 distinct numbers among the remaining elements with sum N .
5. (USAJMO 2016) Let $\triangle ABC$ be an acute triangle, with O as its circumcenter. Point H is the foot of the perpendicular from A to line $\overleftrightarrow{BC}$, and points P and Q are the feet of the perpendiculars from H to the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$, respectively.

Given that

$$AH^2 = 2 \cdot AO^2,$$

prove that the points O , P , and Q are collinear.

6. (*USAJMO 2016*) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real numbers x and y ,

$$(f(x) + xy) \cdot f(x - 3y) + (f(y) + xy) \cdot f(3x - y) = (f(x + y))^2.$$

§8 USAJMO 2017

1. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Prove that there are infinitely many distinct pairs (a, b) of relatively prime positive integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

2. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Consider the equation

$$(3x^3 + xy^2)(x^2y + 3y^3) = (x - y)^7.$$

(a) Prove that there are infinitely many pairs (x, y) of positive integers satisfying the equation.

(b) Describe all pairs (x, y) of positive integers satisfying the equation.

3. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be an equilateral triangle and let P be a point on its circumcircle. Let lines PA and BC intersect at D ; let lines PB and CA intersect at E ; and let lines PC and AB intersect at F . Prove that the area of triangle DEF is twice the area of triangle ABC .

4. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Are there any triples (a, b, c) of positive integers such that $(a - 2)(b - 2)(c - 2) + 12$ is a prime that properly divides the positive number $a^2 + b^2 + c^2 + abc - 2017$?

5. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let O and H be the circumcenter and the orthocenter of an acute triangle ABC . Points M and D lie on side BC such that $BM = CM$ and $\angle BAD = \angle CAD$.

Ray MO intersects the circumcircle of triangle BHC in point N . Prove that $\angle ADO = \angle HAN$.

6. (*USAJMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $P_1, \dots, P_{2n}$ be $2n$ distinct points on the unit circle $x^2 + y^2 = 1$ other than $(1, 0)$. Each point is colored either red or blue, with exactly n of them red and exactly n of them blue. Let $R_1, \dots, R_n$ be any ordering of the red points. Let B_1 be the nearest blue point to R_1 traveling counterclockwise around the circle starting from R_1 . Then let B_2 be the nearest of the remaining blue points to R_2 traveling counterclockwise around the circle from R_2 , and so on, until we have labeled all the blue points $B_1, \dots, B_n$. Show that the number of counterclockwise arcs of the form $R_i \rightarrow B_i$ that contain the point $(1, 0)$ is independent of the way we chose the ordering $R_1, \dots, R_n$ of the red points.

§9 USAJMO 2018

1. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For each positive integer n , find the number of n -digit positive integers that satisfy both of the following conditions:

- no two consecutive digits are equal, and
- the last digit is a prime.

2. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a, b, c be positive real numbers such that $a + b + c = 4\sqrt[3]{abc}$. Prove that

$$2(ab + bc + ca) + 4 \min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

3. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a quadrilateral inscribed in circle ω with $\overline{AC} \perp \overline{BD}$. Let E and F be the reflections of D over lines BA and BC , respectively, and let P be the intersection of lines BD and EF . Suppose that the circumcircle of $\triangle EPD$ meets ω at D and Q , and the circumcircle of $\triangle FPD$ meets ω at D and R . Show that $EQ = FR$.

4. (*USAJMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Triangle ABC is inscribed in a circle of radius 2 with $\angle ABC \geq 90^\circ$, and x is a real number satisfying the equation $x^4 + ax^3 + bx^2 + cx + 1 = 0$, where $a = BC, b = CA, c = AB$. Find all possible values of x .

5. (*USAJMO 2018*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let p be a prime, and let $a_1, \dots, a_p$ be integers. Show that there exists an integer k such that the numbers

$$a_1 + k, a_2 + 2k, \dots, a_p + pk$$

produce at least $\frac{1}{2}p$ distinct remainders upon division by p .

6. (*USAJMO 2018*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Karl starts with n cards labeled $1, 2, 3, \dots, n$ lined up in a random order on his desk. He calls a pair (a, b) of these cards swapped if $a > b$ and the card labeled a is to the left of the card labeled b . For instance, in the sequence of cards $3, 1, 4, 2$, there are three swapped pairs of cards, $(3, 1)$, $(3, 2)$, and $(4, 2)$.

He picks up the card labeled 1 and inserts it back into the sequence in the opposite position: if the card labeled 1 had i card to its left, then it now has i cards to its right. He then picks up the card labeled 2 and reinserts it in the same manner, and so on until he has picked up and put back each of the cards $1, 2, \dots, n$ exactly once in that order. (For example, the process starting at $3, 1, 4, 2$ would be $3, 1, 4, 2 \rightarrow 3, 4, 1, 2 \rightarrow 2, 3, 4, 1 \rightarrow 2, 4, 3, 1 \rightarrow 2, 3, 4, 1$.)

Show that, no matter what lineup of cards Karl started with, his final lineup has the same number of swapped pairs as the starting lineup.

§10 USAJMO 2019

1. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

There are $a + b$ bowls arranged in a row, numbered 1 through $a + b$, where a and b are given positive integers. Initially, each of the first a bowls contains an apple, and each of the last b bowls contains a pear.

A legal move consists of moving an apple from bowl i to bowl $i + 1$ and a pear from bowl j to bowl $j - 1$, provided that the difference $i - j$ is even. We permit multiple fruits in the same bowl at the same time. The goal is to end up with the first b bowls each containing a pear and the last a bowls each containing an apple. Show that this is possible if and only if the product ab is even.

2. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled

diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{Z}$ be the set of all integers. Find all pairs of integers (a, b) for which there exist functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ and $g : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying

$$f(g(x)) = x + a \quad \text{and} \quad g(f(x)) = x + b$$

for all integers x .

3. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

4. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be a triangle with $\angle ABC$ obtuse. The *A-excircle* is a circle in the exterior of $\triangle ABC$ that is tangent to side BC of the triangle and tangent to the extensions of the other two sides. Let E, F be the feet of the altitudes from B and C to lines AC and AB , respectively. Can line EF be tangent to the *A-excircle*?

5. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let n be a nonnegative integer. Determine the number of ways that one can choose $(n+1)^2$ sets $S_{i,j} \subseteq \{1, 2, \dots, 2n\}$, for integers i, j with $0 \leq i, j \leq n$ such that:

- for all $0 \leq i, j \leq n$, the set $S_{i,j}$ has $i + j$ elements; and
- $S_{i,j} \subseteq S_{k,l}$ whenever $0 \leq i \leq k \leq n$ and $0 \leq j \leq l \leq n$

6. (*USAJMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Two rational numbers $\frac{m}{n}$ and $\frac{n}{m}$ are written on a blackboard, where m and n are relatively prime positive integers. At any point, Evan may pick two of the numbers x and y written on the board and write either their arithmetic mean $\frac{x+y}{2}$ or their harmonic mean $\frac{2xy}{x+y}$ on the board as well. Find all pairs (m, n) such that Evan can write 1 on the board in finitely many steps.

§11 USAJMO 2020

1. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled

diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. Carl has n books arranged on a bookshelf. Each book has a height and a width. No two books have the same height, and no two books have the same width. Initially, the books are arranged in increasing order of height from left to right. In a move, Carl picks any two adjacent books where the left book is wider and shorter than the right book, and swaps their locations. Carl does this repeatedly until no further moves are possible. Prove that regardless of how Carl makes his moves, he must stop after a finite number of moves, and when he does stop, the books are sorted in increasing order of width from left to right.

2. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let ω be the incircle of a fixed equilateral triangle ABC . Let ℓ be a variable line that is tangent to ω and meets the interior of segments BC and CA at points P and Q , respectively. A point R is chosen such that $PR = PA$ and $QR = QB$. Find all possible locations of the point R , over all choices of ℓ .

3. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

An empty $2020 \times 2020 \times 2020$ cube is given, and a 2020×2020 grid of square unit cells is drawn on each of its six faces. A beam is a $1 \times 1 \times 2020$ rectangular prism. Several beams are placed inside the cube subject to the following conditions:

- The two 1×1 faces of each beam coincide with unit cells lying on opposite faces of the cube. (Hence, there are $3 \cdot 2020^2$ possible positions for a beam.)
- No two beams have intersecting interiors.
- The interiors of each of the four 1×2020 faces of each beam touch either a face of the cube or the interior of the face of another beam.

What is the smallest positive number of beams that can be placed to satisfy these conditions?

4. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $ABCD$ be a convex quadrilateral inscribed in a circle and satisfying $DA < AB = BC < CD$. Points E and F are chosen on sides CD and AB such that $BE \perp AC$ and $EF \parallel BC$. Prove that $FB = FD$.

5. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Suppose that $(a_1, b_1), (a_2, b_2), \dots, (a_{100}, b_{100})$ are distinct ordered pairs of non-negative integers. Let N denote the number of pairs of integers (i, j) satisfying

$1 \leq i < j \leq 100$ and $|a_i b_j - a_j b_i| = 1$. Determine the largest possible value of N over all possible choices of the 100 ordered pairs.

6. (*USAJMO 2020*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. Let $P(x_1, x_2, \dots, x_n)$ be a nonconstant n -variable polynomial with real coefficients. Assume that whenever $r_1, r_2, \dots, r_n$ are real numbers, at least two of which are equal, we have $P(r_1, r_2, \dots, r_n) = 0$. Prove that $P(x_1, x_2, \dots, x_n)$ cannot be written as the sum of fewer than $n!$ monomials. (A monomial is a polynomial of the form $cx_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$, where c is a nonzero real number and $d_1, d_2, \dots, d_n$ are nonnegative integers.)

§12 USAJMO 2021

1. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for positive integers a and b ,

$$f(a^2 + b^2) = f(a)f(b) \text{ and } f(a^2) = f(a)^2.$$

2. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

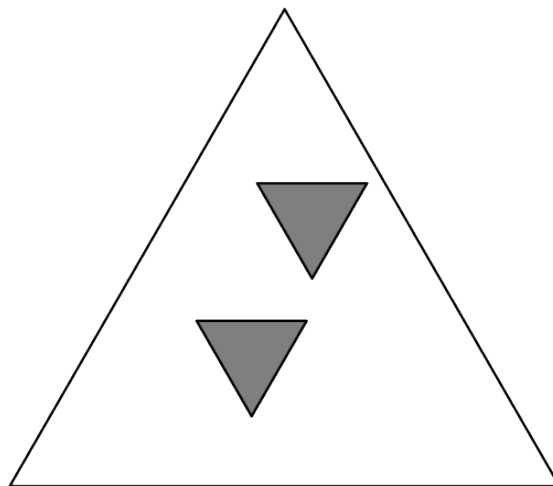
Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

3. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

An equilateral triangle Δ of side length $L > 0$ is given. Suppose that n equilateral triangles with side length 1 and with non-overlapping interiors are drawn inside Δ , such that each unit equilateral triangle has sides parallel to Δ , but with opposite orientation. (An example with $n = 2$ is drawn below.)



Prove that

$$n \leq \frac{2}{3}L^2.$$

4. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Carina has three pins, labeled A, B , and C , respectively, located at the origin of the coordinate plane. In a move, Carina may move a pin to an adjacent lattice point at distance 1 away. What is the least number of moves that Carina can make in order for triangle ABC to have area 2021?

(A lattice point is a point (x, y) in the coordinate plane where x and y are both integers, not necessarily positive.)

5. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

A finite set S of positive integers has the property that, for each $s \in S$, and each positive integer divisor d of s , there exists a unique element $t \in S$ satisfying $\gcd(s, t) = d$. (The elements s and t could be equal.) Given this information, find all possible values for the number of elements of S .

6. (*USAJMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 4$ be an integer. Find all positive real solutions to the following system of $2n$ equations:

$$\begin{aligned} a_1 &= \frac{1}{a_{2n}} + \frac{1}{a_2}, & a_2 &= a_1 + a_3, \\ a_3 &= \frac{1}{a_2} + \frac{1}{a_4}, & a_4 &= a_3 + a_5, \\ a_5 &= \frac{1}{a_4} + \frac{1}{a_6}, & a_6 &= a_5 + a_7, \end{aligned}$$

$$\vdots$$

$$a_{2n-1} = \frac{1}{a_{2n-2}} + \frac{1}{a_{2n}}, \quad a_{2n} = a_{2n-1} + a_1$$

§13 USAJMO 2022

1. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For which positive integers m does there exist an infinite arithmetic sequence of integers $a_1, a_2, \dots$ and an infinite geometric sequence of integers $g_1, g_2, \dots$ satisfying the following properties?

- $a_n - g_n$ is divisible by m for all integers $n > 1$;
- $a_2 - a_1$ is not divisible by m .

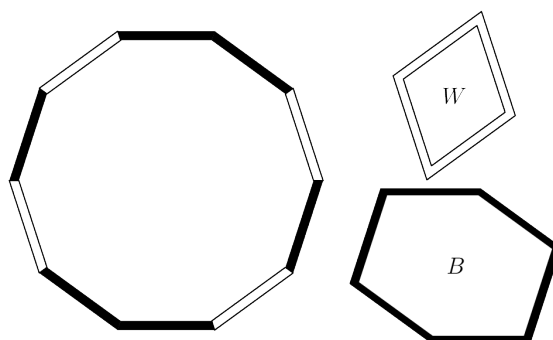
2. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a and b be positive integers. The cells of an $(a + b + 1) \times (a + b + 1)$ grid are colored amber and bronze such that there are at least $a^2 + ab - b$ amber cells and at least $b^2 + ab - a$ bronze cells. Prove that it is possible to choose a amber cells and b bronze cells such that no two of the $a + b$ chosen cells lie in the same row or column.

3. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $b \geq 2$ and $w \geq 2$ be fixed integers, and $n = b + w$. Given are $2b$ identical black rods and $2w$ identical white rods, each of side length 1.

We assemble a regular $2n$ -gon using these rods so that parallel sides are the same color. Then, a convex $2b$ -gon B is formed by translating the black rods, and a convex $2w$ -gon W is formed by translating the white rods. An example of one way of doing the assembly when $b = 3$ and $w = 2$ is shown below, as well as the resulting polygons B and W .



Prove that the difference of the areas of B and W depends only on the numbers b and w , and not on how the $2n$ -gon was assembled.

4. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCD$ be a rhombus, and let K and L be points such that K lies inside the rhombus, L lies outside the rhombus, and $KA = KB = LC = LD$. Prove that there exist points X and Y on lines AC and BD such that $KXLY$ is also a rhombus.

5. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all pairs of primes (p, q) for which $p - q$ and $pq - q$ are both perfect squares.

6. (*USAJMO 2022*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a_0, b_0, c_0 be complex numbers, and define

$$a_{n+1} = a_n^2 + 2b_nc_n$$

$$b_{n+1} = b_n^2 + 2c_na_n$$

$$c_{n+1} = c_n^2 + 2a_nb_n$$

for all nonnegative integers n .

§14 USAJMO 2023

1. (*USAJMO 2023*) Find all triples of positive integers (x, y, z) that satisfy the equation

$$2(x + y + z + 2xyz)^2 = (2xy + 2yz + 2zx + 1)^2 + 2023$$

2. (*USAJMO 2023*) In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.
3. (*USAJMO 2023*) Consider an n -by- n board of unit squares for some odd positive integer n . We say that a collection C of identical dominoes is a maximal grid-aligned configuration on the board if C consists of $(n^2 - 1)/2$ dominoes where each domino covers exactly two neighboring squares and the dominoes don't overlap: C then covers all but one square on the board. We are allowed to slide (but not rotate) a

- domino on the board to cover the uncovered square, resulting in a new maximal grid-aligned configuration with another square uncovered. Let $k(C)$ be the number of distinct maximal grid-aligned configurations obtainable from C by repeatedly sliding dominoes. Find the maximum value of $k(C)$ as a function of n .
4. (*USAJMO 2023*) Two players, B and R , play the following game on an infinite grid of unit squares, all initially colored white. The players take turns starting with B . On B 's turn, B selects one white unit square and colors it blue. On R 's turn, R selects two white unit squares and colors them red. The players alternate until B decides to end the game. At this point, B gets a score, given by the number of unit squares in the largest (in terms of area) simple polygon containing only blue unit squares. What is the largest score B can guarantee?
- (A simple polygon is a polygon (not necessarily convex) that does not intersect itself and has no holes.)
5. (*USAJMO 2023*) A positive integer a is selected, and some positive integers are written on a board. Alice and Bob play the following game. On Alice's turn, she must replace some integer n on the board with $n + a$, and on Bob's turn he must replace some even integer n on the board with $n/2$. Alice goes first and they alternate turns. If on his turn Bob has no valid moves, the game ends.
- After analyzing the integers on the board, Bob realizes that, regardless of what moves Alice makes, he will be able to force the game to end eventually. Show that, in fact, for this value of a and these integers on the board, the game is guaranteed to end regardless of Alice's or Bob's moves.
6. (*USAJMO 2023*) Isosceles triangle ABC , with $AB = AC$, is inscribed in circle ω . Let D be an arbitrary point inside BC such that $BD \neq DC$. Ray AD intersects ω again at E (other than A). Point F (other than E) is chosen on ω such that $\angle DFE = 90^\circ$. Line FE intersects rays AB and AC at points X and Y , respectively. Prove that $\angle XDE = \angle EDY$.

§15 USAJMO 2024

1. (*USAJMO 2024*) Let $ABCD$ be a cyclic quadrilateral with $AB = 7$ and $CD = 8$. Points P and Q are selected on segment AB such that $AP = BQ = 3$. Points R and S are selected on segment CD such that $CR = DS = 2$. Prove that $PQRS$ is a cyclic quadrilateral.
2. (*USAJMO 2024*) Let m and n be positive integers. Let S be the set of integer points (x, y) with $1 \leq x \leq 2m$ and $1 \leq y \leq 2n$. A configuration of mn rectangles is called *happy* if each point in S is a vertex of exactly one rectangle, and all rectangles have sides parallel to the coordinate axes. Prove that the number of *happy* configurations is odd.
3. (*USAJMO 2024*) Let $a(n)$ be the sequence defined by $a(1) = 2$ and $a(n+1) = (a(n))^{n+1} - 1$ for each integer $n \geq 1$. Suppose that $p > 2$ is prime and k is a positive integer. Prove that some term of the sequence $a(n)$ is divisible by p^k .
4. (*USAJMO 2024*) Let $n \geq 3$ be an integer. Rowan and Colin play a game on an $n \times n$ grid of squares, where each square is colored either red or blue. Rowan is allowed to permute the rows of the grid, and Colin is allowed to permute the columns of the grid. A grid coloring is *orderly* if:

- no matter how Rowan permutes the rows of the coloring, Colin can then permute the columns to restore the original grid coloring; and
- no matter how Colin permutes the columns of the coloring, Rowan can then permute the rows to restore the original grid coloring;

In terms of n , how many orderly colorings are there?

5. (USAJMO 2024) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$$

for all $x, y \in \mathbb{R}$.

6. (USAJMO 2024) Point D is selected inside acute triangle ABC so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC . Show that line AB is tangent to the circumcircle of triangle BEM .

§16 USAJMO 2025

1. (USAJMO 2025) Let $\mathbb{Z}$ be the set of integers, and let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be a function. Prove that there are infinitely many integers c such that the function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $g(x) = f(x) + cx$ is not bijective. Note: A function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ is bijective if for every integer b , there exists exactly one integer a such that $g(a) = b$.
2. (USAJMO 2025) Let k and d be positive integers. Prove that there exists a positive integer N such that for every odd integer $n > N$, the digits in the base-2 n representation of n^k are all greater than d .
3. (USAJMO 2025) Let m and n be positive integers, and let $\mathcal{R}$ be a $2m \times 2n$ grid of unit squares.

A domino is a 1×2 or 2×1 rectangle. A subset S of grid squares in $\mathcal{R}$ is domino-tileable if dominoes can be placed to cover every square of S exactly once with no domino extending outside of S . Note: The empty set is domino tileable.

An up-right path is a path from the lower-left corner of $\mathcal{R}$ to the upper-right corner of $\mathcal{R}$ formed by exactly $2m + 2n$ edges of the grid squares.

Determine, with proof, in terms of m and n , the number of up-right paths that divide $\mathcal{R}$ into two domino-tileable subsets.

4. (USAJMO 2025) Let n be a positive integer, and let $a_0, a_1, \dots, a_n$ be nonnegative integers such that $a_0 \geq a_1 \geq \dots \geq a_n$. Prove that

$$\sum_{i=0}^n i \binom{a_i}{2} \leq \frac{1}{2} \binom{a_0 + a_1 + \dots + a_n}{2}.$$

Note: $\binom{k}{2} = \frac{k(k-1)}{2}$ for all nonnegative integers k .

5. (USAJMO 2025) Let H be the orthocenter of acute triangle ABC , let F be the foot of the altitude from C to AB , and let P be the reflection of H across BC . Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y . Prove that C is the midpoint of XY .

6. (*USAJMO 2025*) Let S be a set of integers with the following properties:
- $\{1, 2, \dots, 2025\} \subseteq S$.
 - If $a, b \in S$ and $\gcd(a, b) = 1$, then $ab \in S$.
 - If for some $s \in S$, $s + 1$ is composite, then all positive divisors of $s + 1$ are in S .
- Prove that S contains all positive integers.

§17 USAJMO 2026

1. (*USAJMO 2026*) Let a, b, c be distinct positive integers such that $ab + c = c^2$. Prove that $(a - b)^2 \geq 4c$.
2. (*USAJMO 2026*) There are m foxes and n rabbits sitting in $m + n$ seats around a circular table, where all animals are distinguishable. If a fox and a rabbit are sitting next to each other, they may swap positions with each other. For a fixed starting configuration, determine the number of configurations that can be reached by a sequence of swaps.
Note: Rotations and reflections of a configuration are considered distinct.
3. (*USAJMO 2026*) Let ABC be an acute scalene triangle with no angle equal to 60° . Let ω be the circumcircle of ABC . Let $\triangle_B$ be the equilateral triangle with three vertices on ω , one of which is B . Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B . Let $\triangle_C$ and ℓ_C be defined analogously. Let Y to be the intersection of AC and ℓ_B , and let Z be the intersection of AB and ℓ_C .
Suppose that the circumcircle of AYZ intersects ω at $P \neq A$, BC intersects YZ at D , and PA intersects YZ at E . Prove that $PE = PD$.
4. (*USAJMO 2026*) Triangle $\triangle ABC$ has circumcircle ω and circumcenter O . Lines AO and BC meet at point D . Let X be the A -excenter of $\triangle ABD$ and let Y be the A -excenter of $\triangle ACD$. Prove that if X lies on ω , then Y also lies on ω .
5. (*USAJMO 2026*) A positive integer n is called solitary if, for any nonnegative integers a and b such that $a + b = n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026} .
6. (*USAJMO 2026*) Emily has a red sheet of paper. She draws 2026 circles (not necessarily of equal size) on the piece of paper. She chooses a circle to color black, then cuts the paper around the circumference of all 2026 circles. She then separates the pieces of paper, into at least 2 black pieces and some number of red pieces. Is it possible that all black pieces are congruent?